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Mahindra University Hyderabad

École Centrale School of Engineering

Minor-II Examination

Program: B. Tech. **Branch: CM** **Year: II** **Semester: I**

Subject: Signals and Systems (MA2105)

Date: 31/10/2025 **Start Time: 10:00 AM**

Time Duration: 1.5 Hours **Max. Marks: 15**

Instructions:

1) Answer all the questions.
2) All questions are self-explanatory; no clarification will be provided during the exam.

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1. (8 points) Choose the **correct option** for each question and **justify your answer briefly**.

$e^{-as}F(s) = u(t-a)f(t-a)$

$\delta(t-a) = e^{-as}$

(a) The Laplace transform of $e^{-2t}u(t)$ is:

(i) $\frac{1}{s+2}$ (ii) $\frac{1}{s-2}$

(iii) $\frac{s}{s+2}$ (iv) $\frac{s-2}{s^2+4}$

(b) The region of convergence (ROC) of $\mathcal{L}\{e^{3t}u(t)\}$ is:

(i) $\Re(s) < 3$ (ii) $\Re(s) > 3$

(iii) $\Re(s) = 3$ (iv) $\Re(s) \leq 3$

Here $\Re(s)$ denotes the real part of s.

(c) The inverse Laplace transform of

$$\frac{1}{(s+2)(s+3)}.$$

is:

(i) $e^{-2t} - e^{-3t}$ (ii) $e^{-2t} - e^{-3t}/t$

(iii) $e^{-3t} - e^{-2t}$ (iv) $2e^{-2t} + 3e^{-3t}$

(d) Determine A, B so that

$$\frac{2s+1}{(s+2)(s+3)} = \frac{A}{s+2} + \frac{B}{s+3}.$$

(i) $A = -3,\ B = 5$ (ii) $A = 3,\ B = -5$

(iii) $A = 2,\ B = -1$ (iv) $A = 1,\ B = 2$

(e) Which of the following signals are orthogonal over $[0, 2\pi]$?

(i) $\sin t$ and $\sin 2t$ (ii) $\sin t$ and $\cos t$

(iii) $\sin 2t$ and $\sin 2t$ (iv) $\cos t$ and $\cos t$

P.T.O.

(f) For the differential equation

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = 5x(t),$$

find the transfer function $H(s) = \frac{Y(s)}{X(s)}$.

(i) $\frac{5}{s^2+3s+2}$ (ii) $\frac{2}{s^2+5s+3}$

(iii) $\frac{s+3}{5s^2+2s}$ (iv) $\frac{5s}{s^2+2s+3}$

(g) Evaluate the following limit and identify the resulting signal:

$$\lim_{a\to 0}\left[e^{-at}u(t) - e^{at}u(-t)\right].$$

(i) Unit step signal $u(t)$ (ii) Ramp signal $r(t)$

(iii) Signum signal $\mathrm{sgn}(t)$ (iv) None of these

(h) The Fourier transform of $e^{-a|t|u(t)}$ is :

(i) $\frac{2a}{a^2+\omega^2}$ (ii) $\frac{a}{a^2+\omega^2}$

(iii) $\frac{1}{a+j\omega}$ (iv) None of these

Here, $u(t)$ denotes the unit step signal.

2. (3 points) Find:

$$\mathcal{L}^{-1}\left\{\frac{s^2+2s+5}{s(s+1)(s^2+4)}\right\}.$$

or

Solve:

$$y''' + y'' - y' - y = e^{2t}, \quad y(0) = 0,\ y'(0) = 1,\ y''(0) = 0.$$

3. (4 points) Evaluate the trigonometric Fourier series expansion of a signal $x(t) = |\cos(t)|$.

or

Evaluate the exponential Fourier series coefficients of the periodic continuous-time signal

$$x(t) = 1 + \sin(\omega_0 t) + 2\cos(\omega_0 t) + \cos\left(2\omega_0 t + \frac{\pi}{4}\right)$$